

CS 486/686

Uninformed Search

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Lecture 2

RN 3.1–3.4

Search ›

Uncertainty ›

Decisions ›

Learning

Learning goals

- Formulate problems as search
- Trace DFS, BFS, IDS
- Analyze: space, time, completeness, optimality
- Pick the right algorithm

Heads-up: Chat 1 is live

Sign up: andromeda-208.cs.uwaterloo.ca

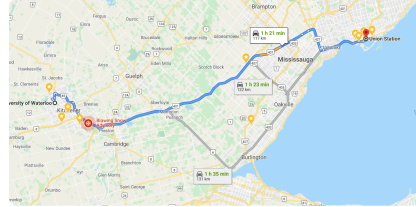
- Use your **WatIAM @uwaterloo.ca** email (not your friendly one)
- Verify the email Chrysalis sends; you'll be auto-added to CS 486/686
- **Chat 1** covers L1 + L2 — due **Tue May 26, 11:59 PM** (*one-time extension for late enrollees*)
- Issues? Private Piazza post to Prashanth Arun & Robert Cai (Chrysalis designers)

Full instructions: [course page → Chat Assignments with Chrysalis](#).

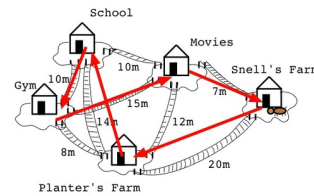
Where does search show up?



Robotics



Route planning



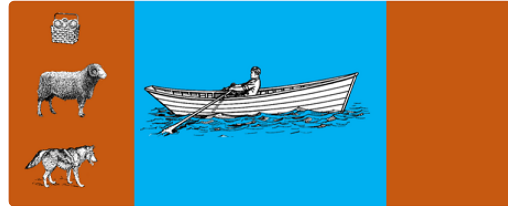
TSP

$$((a \wedge b) \vee c) \wedge d \\ \vee (\neg e)$$

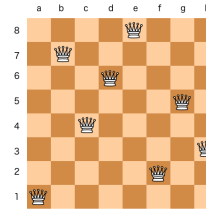
SAT



8-puzzle



River crossing



N-queens

...

And many more

Why search?

- Sequential decisions
- No closed-form algorithm
- Easy to verify, hard to find
- Systematic, not guesswork

What is a search problem?

Five components:

- States
- Initial state
- Goal test
- Successor function
- Cost (optional)

Solution = path from start to goal.

State spaces are huge

1.8×10^5

8-puzzle

3×3 sliding tile

10^{13}

15-puzzle

4×4 sliding tile

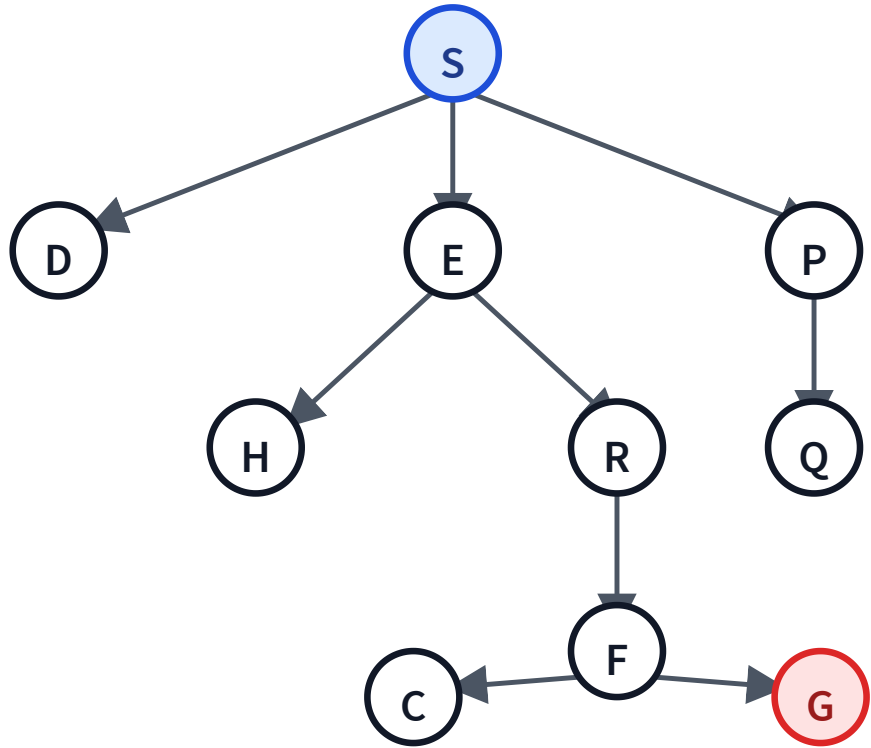
4.3×10^{19}

Rubik's cube

3×3×3

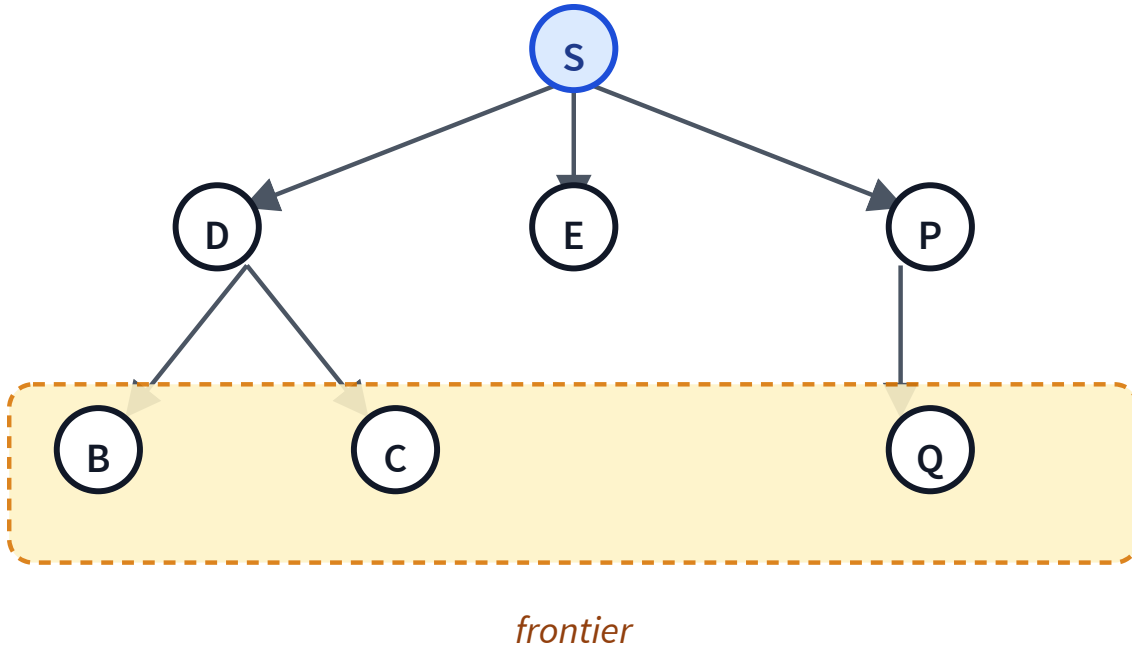
Too many states to enumerate. We need **clever search**.

Search graph



- States → **nodes**
- Actions → **edges**
- Goal: find a path **S** → **G**

Search tree & frontier



- Don't enumerate the graph
- Grow a tree from S, one path at a time
- **Frontier** = paths to expand next

Example: 8-puzzle

Initial			Goal		
5	3		1	2	3
8	7	6	4	5	6
2	4	1	7	8	

- **State:** 9 positions, 0 = empty
- **Initial:** 530, 876, 241
- **Goal:** 123, 456, 780
- **Successor:** slide adjacent tile into empty
- **Cost:** 1 per move

Quick check: successors

Q. Which is a successor of 530,876,241?

- A. 350,876,241
- B. 536,870,241
- C. 537,806,241
- D. 538,076,241

Initial

5	3	
8	7	6
2	4	1

B. Slide the 6 up.

Multiple formulations possible

- State def $\rightarrow$ nodes
- Successor $\rightarrow$ edges
- Minimize both

Same 8-puzzle, different state

By grid

5	3	
8	7	6
2	4	1

9 cells, 0–8

VS

By coordinates

$(x_0, y_0), (x_1, y_1),$
 $(x_2, y_2), \dots, (x_8, y_8)$

9 (x,y) pairs — ~2× the space

Successor swaps empty tile's coords with a neighbour's.

Generic search algorithm

```
// Inputs: graph, start node s, goal test goal(n)  
function Search(graph, s, goal):  
    frontier  $\leftarrow$  { <s> }  
    while frontier is not empty:  
        select and remove a path < $n_0, \dots, n_k$ > from frontier  
        if goal( $n_k$ ):  
            return < $n_0, \dots, n_k$ >  
        for each neighbour n of  $n_k$ :  
            add < $n_0, \dots, n_k, n$ > to frontier  
    return "no solution"
```

expand:

pop path



get neighbours



push extended paths

One template, three algorithms

DFS, BFS, IDS all use this skeleton.

They differ only in **which path** they pull from the frontier.

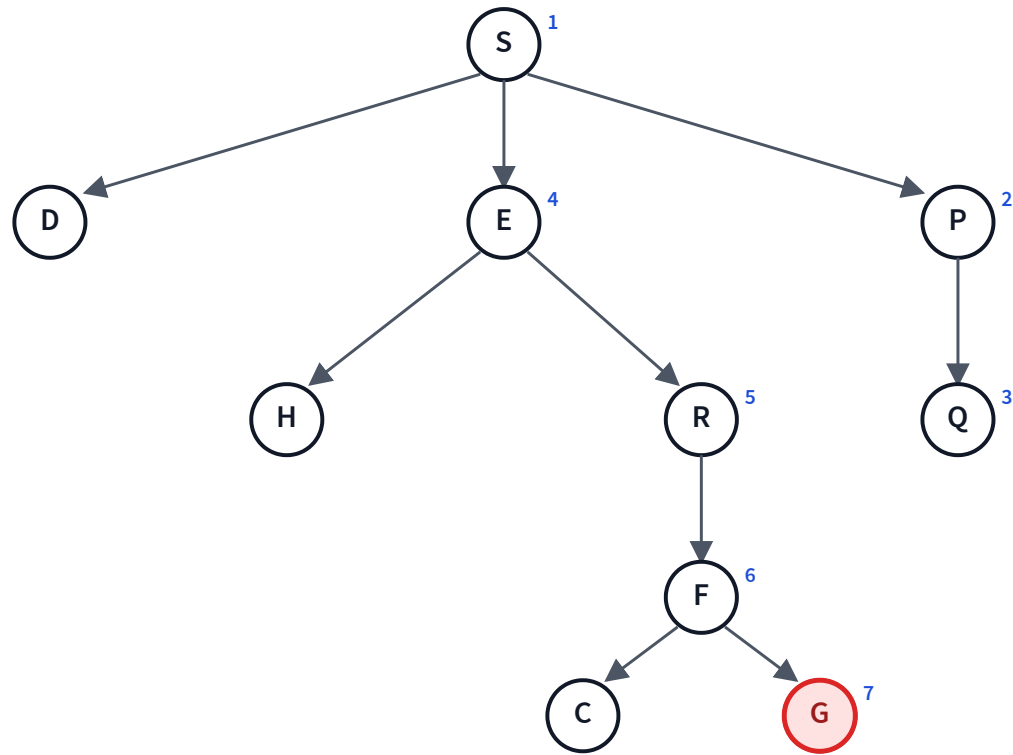
Depth-first search (DFS)

- Frontier = **stack** (LIFO)
- Pop newest
- Dive deep, then backtrack

Trace: DFS

Push children alphabetically. Step numbers = order of expansion.

Frontier:

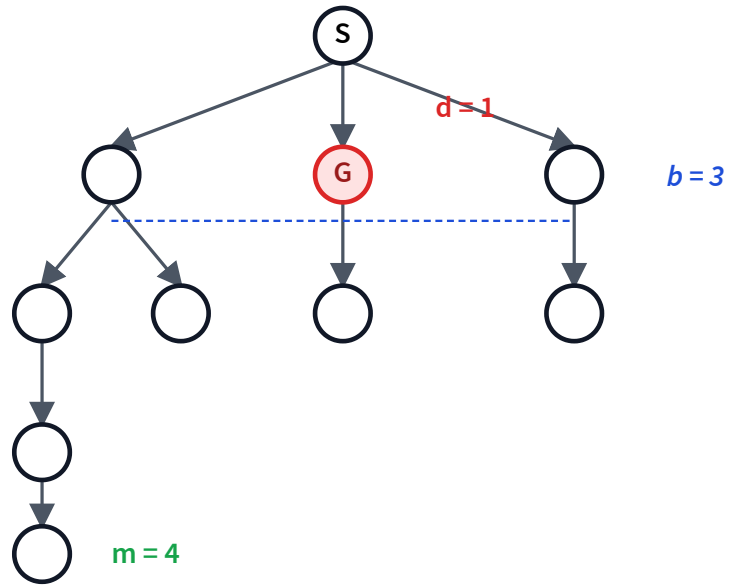


Evaluating search algorithms

Four questions we'll ask about every algorithm:

- **Space** — how much memory? (worst-case frontier size)
- **Time** — how many nodes does it visit? (worst case)
- **Complete?** — is it guaranteed to find a goal if one exists?
- **Optimal?** — does it return the cheapest (or shallowest) goal?

Tree parameters: b, m, d



b — branching factor

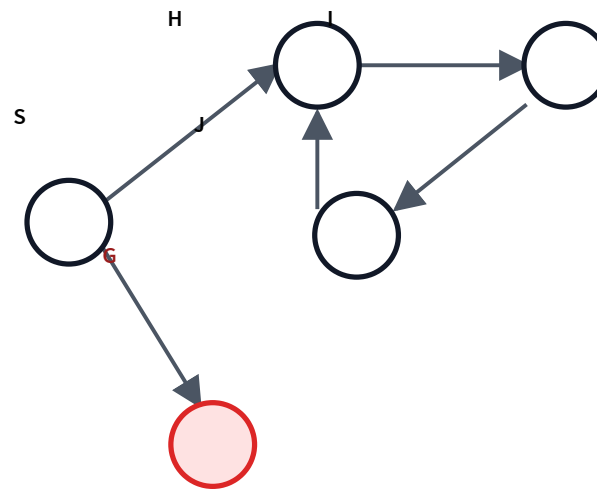
d — depth of shallowest goal

m — max depth of tree

DFS — properties

Space	$O(bm)$ <i>linear</i>
Time	$O(b^m)$ <i>exponential</i>
Complete?	No
Optimal?	No

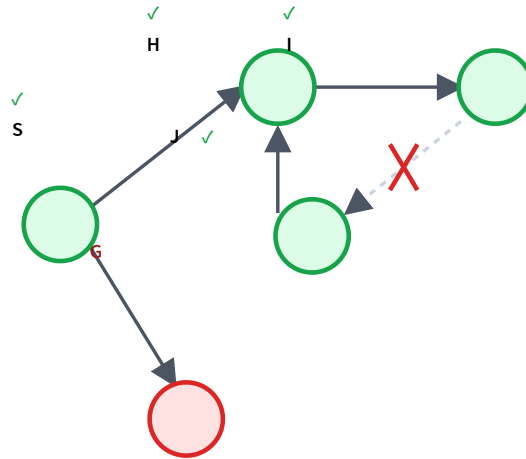
DFS fails on cycles



$H \rightarrow I \rightarrow J \rightarrow H$ traps DFS forever.

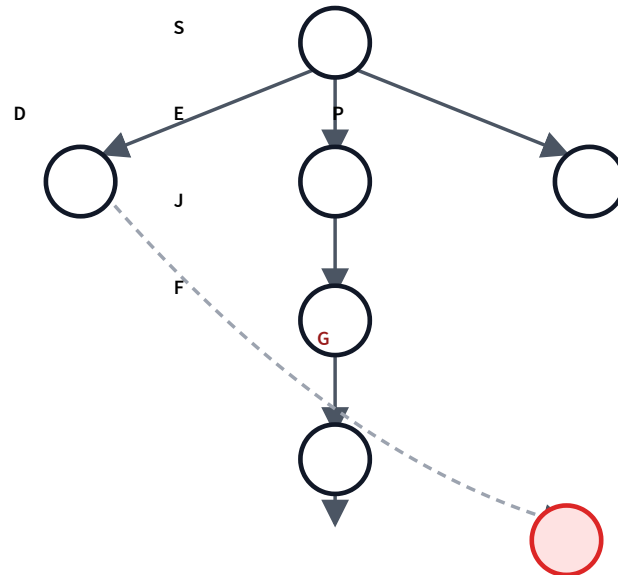
Fix: cycle pruning

Keep a **visited** set. Skip states we've already expanded.



Order: $S \rightarrow H \rightarrow I \rightarrow J \rightarrow \textit{back to H? skip!} \rightarrow G$.

DFS takes the first goal, not the best



Returns the long path; ignores the shorter $S \rightarrow D \dashrightarrow G$.

When to use DFS

Good fit

- Memory limited
- Many goals exist
- Deep solutions

Bad fit

- Cycles
- Shallow solutions
- Need optimality

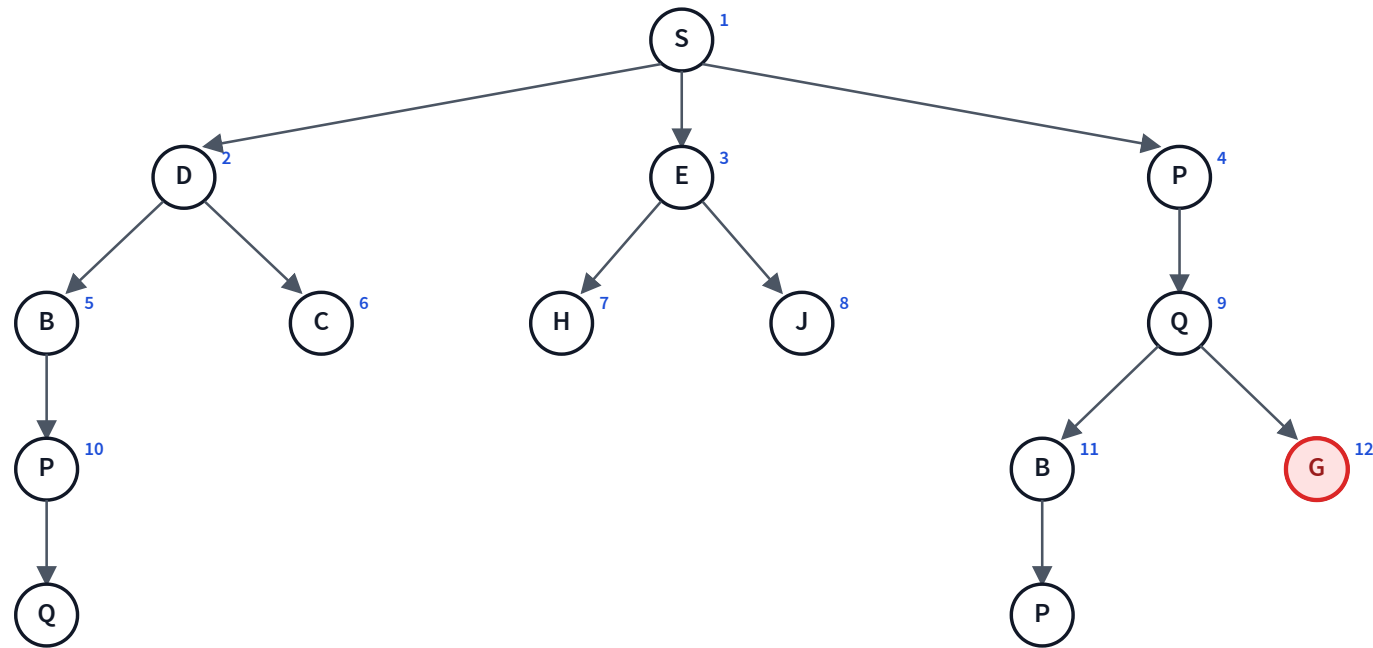
Breadth-first search (BFS)

- Frontier = **queue** (FIFO)
- Pop oldest
- Level by level

Trace: BFS

Pop the oldest path. Step number = order of expansion. Multiple paths to *P* and *B*.

Frontier:



BFS — properties

Space	$O(b^d)$ <i>exponential</i>
Time	$O(b^d)$ <i>exponential</i>
Complete?	Yes
Optimal?	Shallowest goal (<i>optimal at equal cost</i>)

When to use BFS

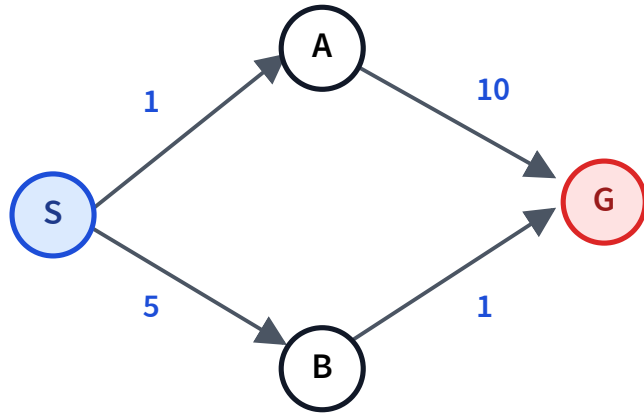
Good fit

- Memory plentiful
- Want fewest arcs
- Cycles possible

Bad fit

- Deep solutions
- Large branching
- Graph generated on the fly

What if edges have different costs?



$S \rightarrow A \rightarrow G$: cost 11. $S \rightarrow B \rightarrow G$: cost 6.

BFS picks the wrong one (it's fewer hops, but pricier).

Uniform-cost search (UCS)

- Frontier = **priority queue**
- Key = path cost so far
- Pop the **cheapest** path first

Complete + optimal when all costs are positive.

When every edge has cost 1, UCS = BFS. (We'll use BFS in this lecture for that reason.)

BFS vs DFS — Q1

Q. Memory is very limited.

- A. BFS
- B. DFS
- C. Both are fine
- D. Neither

B — DFS. DFS: $O(bm)$ vs BFS: $O(b^d)$.

BFS vs DFS — Q2

Q. All solutions are deep in the search tree.

- A. BFS
- B. DFS
- C. Both are fine
- D. Neither

B — DFS. Dives down long paths instead of enumerating every shallower level.

BFS vs DFS — Q3

Q. The graph contains cycles.

- A. BFS
- B. DFS
- C. Both are fine
- D. Neither

A — BFS. DFS can *fail to find* the goal (gets stuck going deeper into the cycle); BFS still finds any goal at finite depth. (BFS does waste work without cycle pruning — but it's complete.)

BFS vs DFS — Q4

Q. The branching factor is huge or infinite.

- A. BFS
- B. DFS
- C. Both are fine
- D. Neither

B — DFS. BFS's frontier blows up with b ; DFS stays linear.

BFS vs DFS — Q5

Q. You must find the *shallowest* goal.

- A. BFS
- B. DFS
- C. Both are fine
- D. Neither

A — BFS. BFS guarantees the shallowest goal.

BFS vs DFS — Q6

Q. All solutions are very shallow.

- A. BFS
- B. DFS
- C. Both are fine
- D. Neither

A — BFS. Checks shallow neighbours before diving deep.

BFS vs DFS tradeoff

	BFS	DFS
Space	$O(b^d)$	$O(bm)$
Complete?	Yes	No

Can we have **both**?

Why $O(b^d)$ matters

Branching factor $b = 10$, 1 state = 8 bytes, computer doing 10^9 states/sec.

Depth d	States b^d	BFS memory	Wall-clock time
4	10^4	80 KB	10 μ s
8	10^8	800 MB	0.1 s
12	10^{12}	8 TB	~17 min
16	10^{16}	80 PB	~4 months
20	10^{20}	800 EB	~3000 years

Doubling d doesn't make the problem twice as hard. It makes it 10,000 $\times$ harder.

Iterative-Deepening Search (IDS)

For $\ell = 1, 2, 3, \dots$, run DFS up to depth ℓ .

Stop when a goal is found.

Each iteration is depth-limited search (DLS).

Trace: IDS

Same graph as BFS. Each iteration is depth-limited DFS.

Depth limit:

IDS — properties

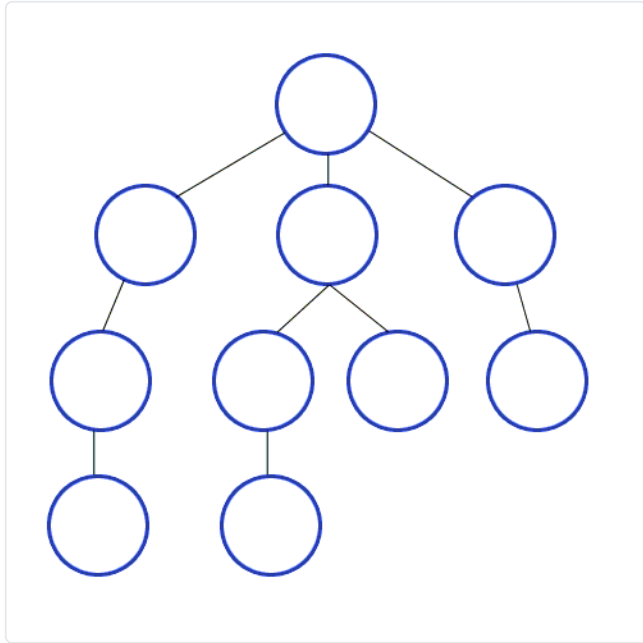
Space	$O(bd)$ <i>linear</i>
Time	$O(b^d)$ <i>exponential</i>
Complete?	Yes
Optimal?	Shallowest goal

Summary: DFS vs BFS vs IDS

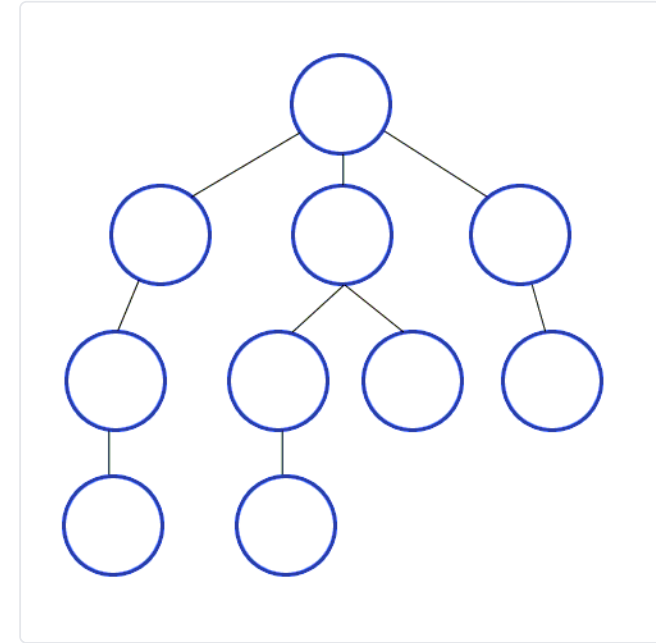
	DFS	BFS	IDS
Frontier	Stack (LIFO)	Queue (FIFO)	Stack, repeated at deeper limits
Space	$O(bm)$	$O(b^d)$	$O(bd)$
Time	$O(b^m)$	$O(b^d)$	$O(b^d)$
Complete?	No	Yes	Yes
Optimal?	No	Shallowest goal	Shallowest goal

b = branching factor; m = max depth of the search tree; d = depth of the shallowest goal.

DFS vs BFS: exploration order



DFS — dives deep first



BFS — expands by level

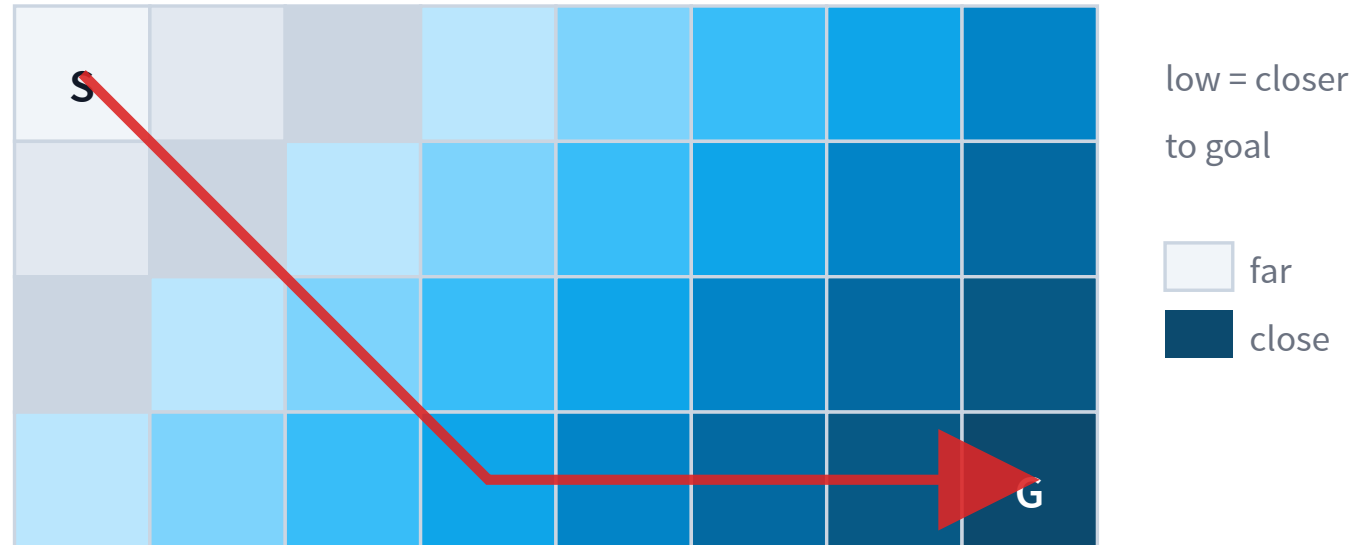
Learning goals (recap)

- ✓ Formulate problems as search
- ✓ Trace DFS, BFS, IDS
- ✓ Analyze: space, time, completeness, optimality
- ✓ Pick the right algorithm

Next lecture: informed search (heuristics, A^).*

Next: informed search

Use a **heuristic** $h(n)$ — an estimate of distance to the goal — to bias the search.



A^* combines path cost so far ($g(n)$) with heuristic ($h(n)$) — provably optimal under mild conditions.